Queue
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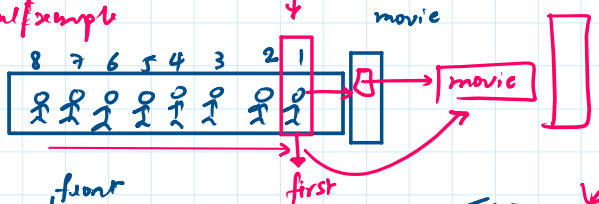
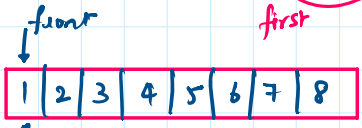
* linear ds
* FIFO: principle → First in first out.
  Stack (ds) → LIFO: Last in first out

: Real example



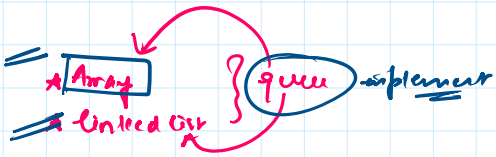
Two pointer:

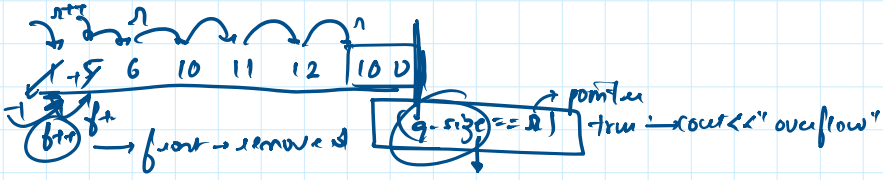


queue<int>q;

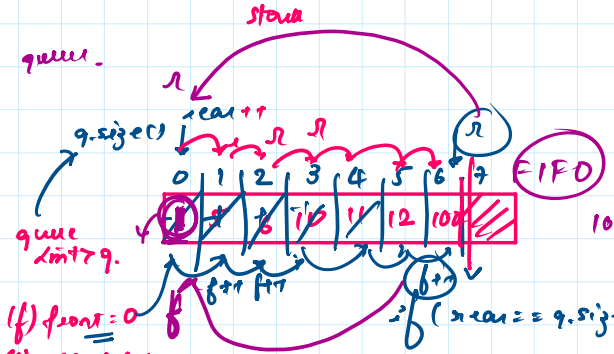
* operations
  pop() → dequeue();
  push() → enqueue();
  top() → front();
  → queue

Stack



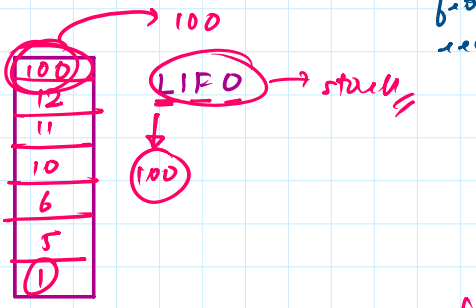


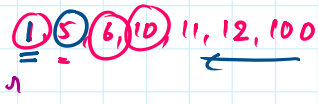
queue.

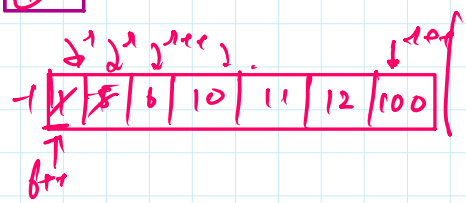


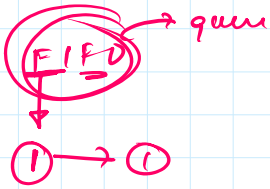
dequeue(), pop()

if(rear==front)
  front = 0
  rear = 0



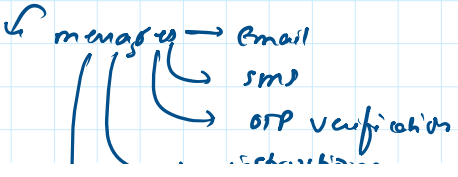


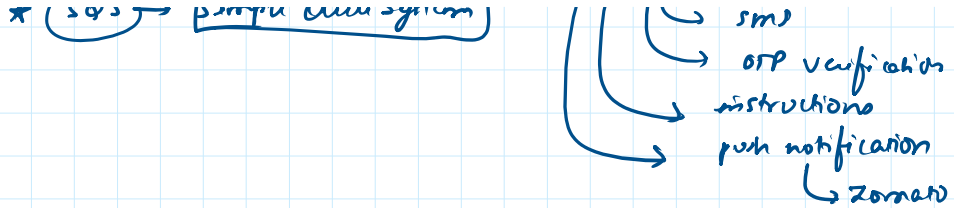


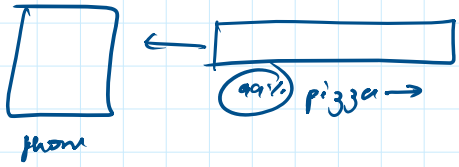


Queue → AWS

* SQS → Simple Queue System







SQS, priority Queue, Queue GitHub

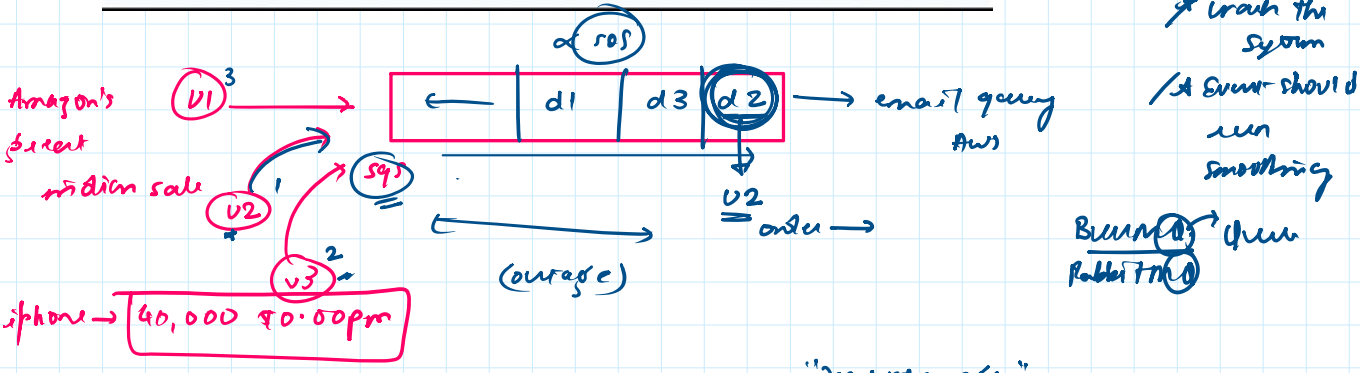
infra sender, OTP

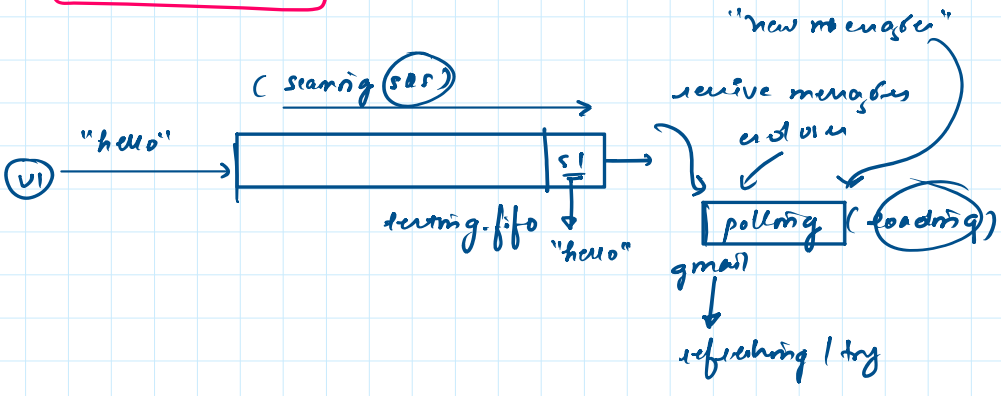
people end user

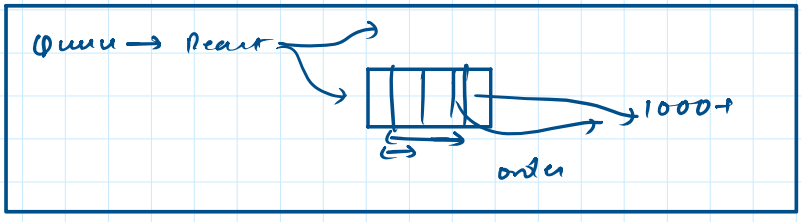
System Design

S/R
* latency
* throughput
* crash the system

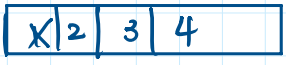
A Event should run smoothly







queue FIFO



pop → dequeue()
= (1)

stack LIFO

pop()
4
= (4)

pop → dequeue()

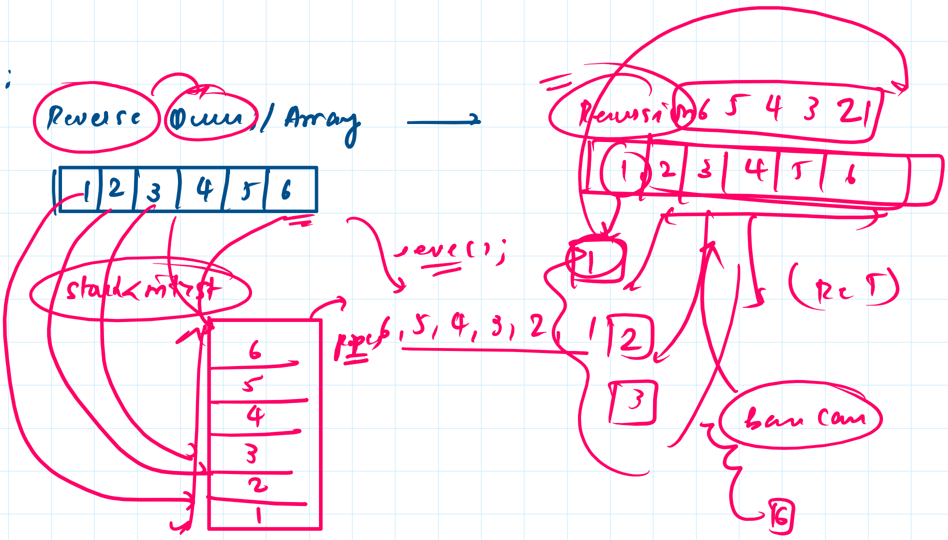
= 1

| 1 |

= 4

Queue:

Reverse Queue / Array →



circular Queue; ; priority Queue "%"

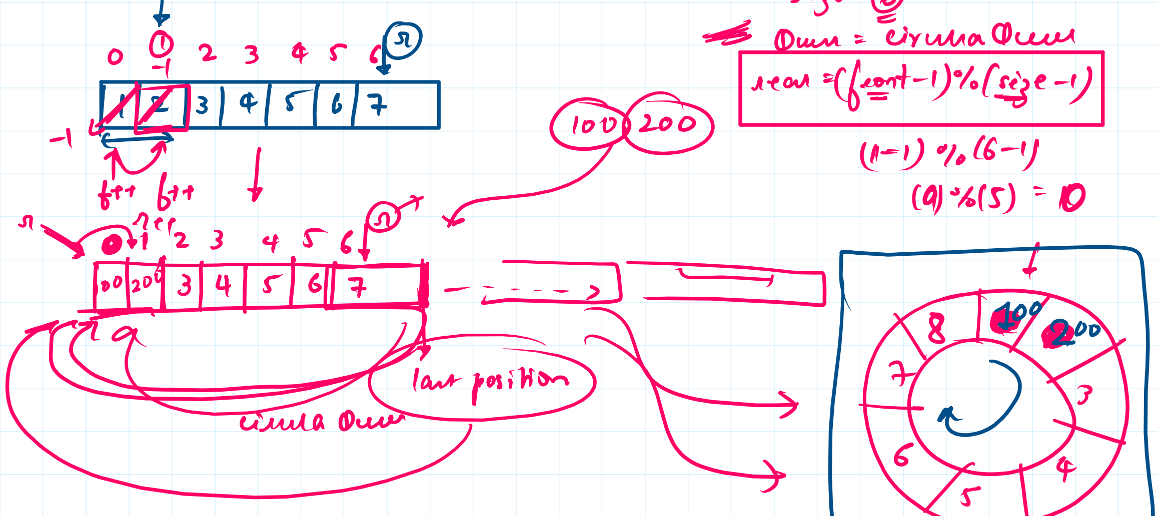
size = 6

Queue = circular Queue

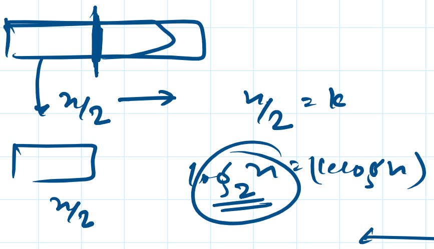
rear = (front − 1) % (size − 1)

(1−1) % (6−1)

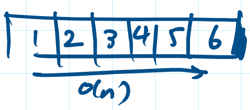
(0) % (5) = 0



O(log n) → binary search, merge sort



arr[]

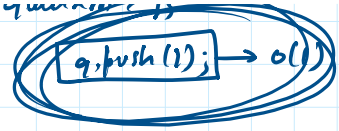


queue<int> q;

q.push(1); → O(1)

for (int i = 0; i < n) {

arr[i+1] = 5]



for (int i = 0; i < n; i++) {

arr[i+m] = 5;